Marburg University

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Exercise Sheet 5 for Advanced Algorithms (Winter 2026/27)

Hand In: *Until 2026-11-20 18:00, on ILIAS.*

Problem 1

20 points

Prove or disprove: p-Clique is W[1]-complete.

Problem 2

40 points

We define the problem RegularIndependentSet problem as the subset of the IndependentSet problem on regular graphs, i.e., graphs where all vertices have the same degree.

Prove that p-IndependentSet $\leq_{fpt}$ p-RegularIndependentSet.